

What is Deep Learning?

What is deep learning in AI?

Deep learning is an [artificial intelligence \(AI\)](#) method that teaches computers to process data in a way inspired by the human brain. Deep learning models can recognize complex pictures, text, sounds, and other data patterns to produce accurate insights and predictions. You can use deep learning methods to automate tasks that typically require human intelligence, such as describing images or transcribing a sound file into text.

What is deep generative learning?

Deep generative learning is deep learning that focuses on creating new output from learned input. Traditionally, deep learning focused on identifying relationships between data. Deep learning models were trained with large amounts of data to recognize patterns in the data set.

Deep generative learning adds generation to pattern recognition. Such models look for data patterns and then create their own unique patterns. For example, they can analyze the text in several books and then use the information to generate new sentences and paragraphs not found in the original books.

Deep generative learning is the basis of modern [generative AI](#) and [foundation models](#). These models use deep learning technologies at scale, trained on vast data, to perform complex tasks like answering questions, creating images from text, and writing content.

Why is deep learning important?

Deep learning technology drives many artificial intelligence applications used in everyday products, such as the following:

- Chatbots and code generators
- Digital assistants
- Voice-activated television remotes
- Fraud detection
- Automatic facial recognition

It is also a critical component of technologies like self-driving cars, virtual reality, and more. Businesses use deep learning models to analyze data and make predictions in various applications.

What are deep learning use cases?

Deep learning has several use cases in automotive, aerospace, manufacturing, electronics, medical research, and other fields.

- Self-driving cars use deep learning models for object detection.
- Defense systems use deep learning to flag areas of interest in satellite images.
- Medical image analysis uses deep learning to detect cancer cells for medical diagnosis.
- Factories use deep learning applications to detect when people or objects are within an unsafe distance of machines.

These various use cases of deep learning can be grouped into five broad categories: computer vision, speech recognition, natural language processing (NLP), recommendation engines, and generative AI.

Computer vision

[Computer vision](#) automatically extracts information and insights from images and videos.

Deep learning techniques to comprehend images in the same way that humans do.

Computer vision has several applications, such as the following:

- Content moderation to automatically remove unsafe or inappropriate content from image and video archives
- Facial recognition to identify faces and recognize attributes like open eyes, glasses, and facial hair
- Image classification to identify brand logos, clothing, safety gear, and other image details

Speech recognition

Deep learning models can analyze human speech despite varying speech patterns, pitch, tone, language, and accent. Virtual assistants such as Amazon Alexa, [text-to-speech](#), and [speech-to-text software](#) use speech recognition to do the following tasks:

- Assist call center agents and automatically classify calls.
- Convert clinical conversations into documentation in real-time.
- Accurately subtitle videos and meeting recordings for a wider content reach.
- Convert scripts to prompts for intelligent voice assistance.

Natural language processing

Computers use deep learning algorithms to gather insights and [meaning from text data](#) and documents. This ability to process natural, human-created text has several use cases, including:

- Automated virtual agents and [chatbots](#)
- Automatic summarization of documents or news articles
- Business intelligence analysis of long-form documents, such as emails and forms
- Indexing of key phrases that indicate sentiment, such as positive and negative comments on social media

Recommendation engines

Applications can use deep learning methods to track user activity and develop [personalized recommendations](#). They can analyze users' behavior and help them discover new products or services. For example,

- Recommend personalized videos and content.
- Recommend customized products and services.
- Filter search results to highlight relevant content based on user location and behavior

Generative AI

Generative AI applications can create new content and communicate with end users more sophisticatedly. They can assist in automating complex workflows, brainstorming ideas, and intelligent knowledge searches. For example, with generative AI tools like [Amazon Q Business](#) and [Amazon Q Developer](#), users can

- Ask natural language questions and get summarized answers from multiple internal knowledge sources.
- Get code suggestions and automatic code scanning and upgrades.
- Create new documents, emails, and other marketing content faster.

How does deep learning work?

Deep learning models are neural networks designed after the human brain. A human brain contains millions of interconnected biological neurons that work together to learn and process information. Similarly, artificial neurons are software modules called nodes that use mathematical calculations to process data. Deep learning neural networks, or artificial neural networks, comprise many layers of artificial neurons that work together to solve complex problems.

The components of a deep neural network are the following.

Input layer

An artificial neural network has several nodes that input data into it. These nodes make up the system's input layer.

Hidden layer

The input layer processes and passes the data to layers further in the neural network. These hidden layers process information at different levels, adapting their behavior as they receive new information. Deep learning networks have hundreds of hidden layers that they can use to analyze a problem from several different angles.

For example, if you were given an image of an unknown animal that you had to classify, you would compare it with animals you already know. For example, you would look at the shape of its eyes and ears, size, number of legs, and fur pattern. You would try to identify patterns, such as the following:

- The animal has hooves, so it could be a cow or deer.
- The animal has cat eyes, so it could be a wild cat.

The hidden layers in deep neural networks work in the same way. If a deep learning algorithm tries to classify an animal image, each of its hidden layers processes a different animal feature and tries to categorize it accurately.

Output layer

The output layer consists of the nodes that output the data. Deep learning models that output "yes" or "no" answers have only two nodes in the output layer. On the other hand, those that output a wider range of answers have more nodes. Generative AI has a sophisticated output layer to generate new data that matches patterns in its training data set.

What is the difference between machine learning, deep learning, and generative AI?

The terms *machine learning*, *deep learning*, and *generative AI* indicate a progression in neural network technology.

Machine learning

Deep learning is a subset of [machine learning](#). Deep learning algorithms emerged to make traditional machine learning techniques more efficient. Traditional machine learning methods require significant human effort to train the software. For example, in animal image recognition, you need to do the following:

- Manually label hundreds of thousands of animal images.

- Make the machine learning algorithms process those images.
- Test those algorithms on a set of unknown images.
- Identify why some results are inaccurate.
- Improve the dataset by labeling new images to improve result accuracy.

This process is called supervised learning. In supervised learning, result accuracy improves only with a broad and sufficiently varied dataset. For instance, the algorithm might accurately identify black cats but not white cats because the training dataset had more images of black cats. In that case, you would need more labeled data of white cat images to train the machine learning models again.

Benefits of deep learning over machine learning

A deep learning network has the following benefits over traditional machine learning.

Efficient processing of unstructured data

Machine learning methods find unstructured data, such as text documents, challenging to process because the training dataset can have infinite variations. On the other hand, deep learning models can comprehend unstructured data and make general observations without manual feature extraction. For instance, a neural network can recognize that these two different input sentences have the same meaning:

- Can you tell me how to make the payment?
- How do I transfer money?

Hidden relationships and pattern discovery

A deep learning application can analyze large amounts of data more deeply and reveal new insights for which it might not have been trained. For example, consider a deep learning model trained to analyze consumer purchases. The model has data only for the items you have already purchased. However, the artificial neural network can suggest new items you haven't bought by comparing your buying patterns to those of similar customers.

Unsupervised learning

Deep learning models can learn and improve over time based on user behavior. They do not require large variations of labeled datasets. For example, consider a neural network that automatically corrects or suggests words by analyzing your typing behavior. Let's assume it was trained in English and can spell-check English words. However, if you frequently type non-English words, such as *danke*, the neural network automatically learns and autocorrects these words too.

Volatile data processing

Volatile datasets have large variations. One example is loan repayment amounts in a bank. A deep learning neural network can categorize and sort that data by analyzing financial transactions and flagging some for fraud detection.

Generative AI

Generative AI took the neural networks of machine learning and deep learning to the next level. While machine learning and deep learning focus on prediction and pattern recognition, generative AI produces unique outputs based on the patterns it detects. Generative AI technology is built on [transformer architecture](#) that combines several different neural networks to combine data patterns in unique ways. Deep learning networks first convert text, images, and other data into mathematical abstractions and then reconvert them into meaningful new patterns.

What are the challenges of deep learning?

Challenges in implementing deep learning and generative AI are given below.

Large quantities of high-quality data

When you train them on large amounts of high-quality data, deep learning algorithms give better results. Outliers or mistakes in your input dataset can significantly affect the deep learning process. For instance, in our animal image example, the deep learning model might classify an airplane as a turtle if the dataset accidentally introduces non-animal images.

To avoid such inaccuracies, you must clean and process large amounts of data before training deep learning models. The input data preprocessing requires large amounts of data storage capacity.

Large processing power

Deep learning algorithms are compute-intensive and require infrastructure with sufficient compute capacity to function properly. Otherwise, they take a long time to process results.

What are the benefits of generative AI and deep learning in the cloud?

Running generative AI and deep learning on cloud infrastructure helps you design, develop, and train applications faster.

Speed

You can train generative AI and deep learning models faster by using clusters of GPUs and CPUs to perform the complex mathematical operations that your neural networks require. You can then deploy these models to process large amounts of data and produce increasingly relevant results.

Scalability

With the wide range of on-demand resources available through the cloud, you can access virtually unlimited hardware resources to tackle AI deep learning models of any size. Your neural networks can take advantage of multiple processors to seamlessly and efficiently distribute workloads across different processor types and quantities.

Tools

You can access AI and deep learning tools like notebooks, debuggers, profilers, pipelines, AIOps, and more. You can work with existing generative AI models from within the cloud as a service without requiring infrastructure to host the model. Teams can start with generative AI and deep learning applications even with limited knowledge and training.